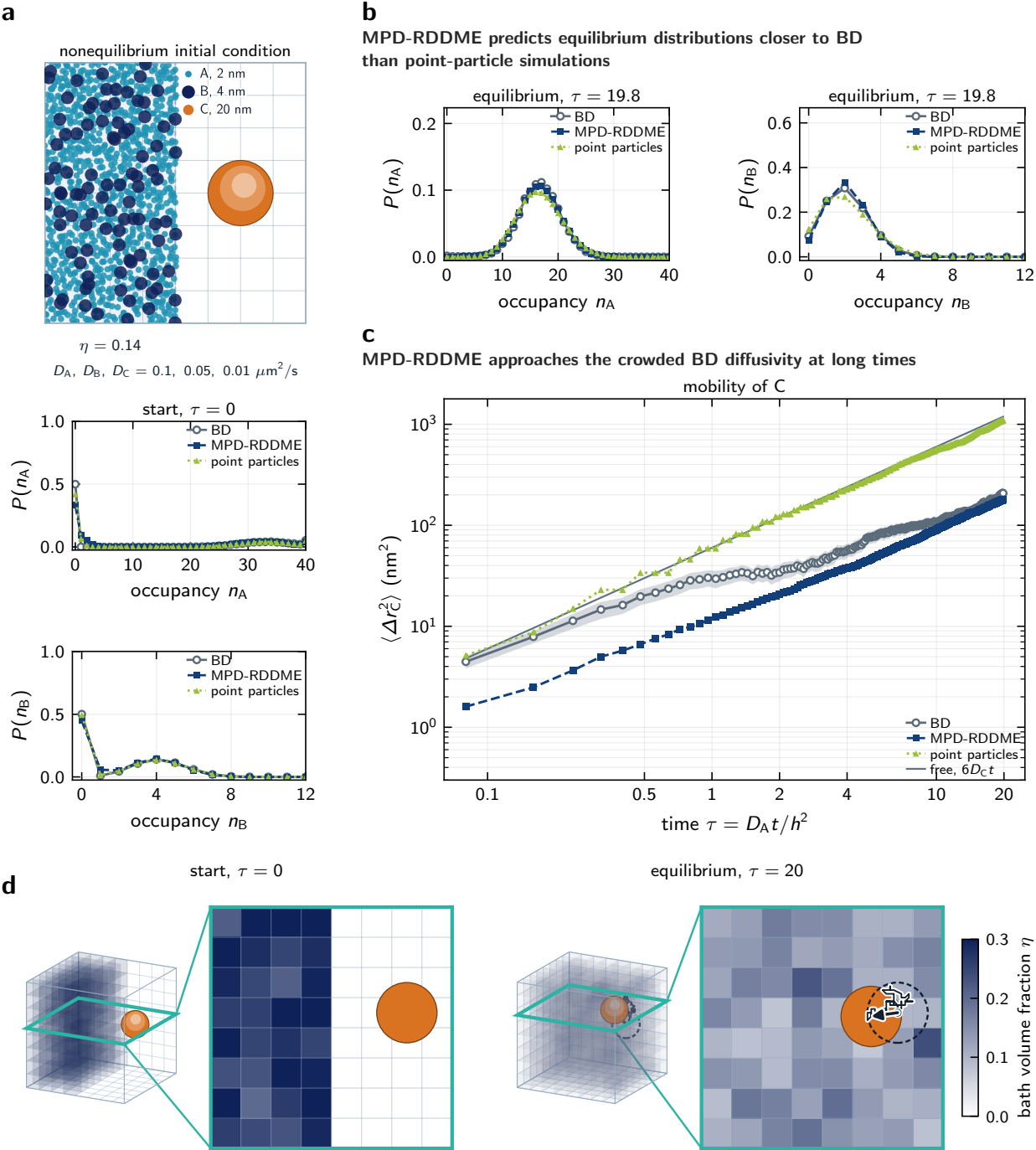

a
nonequilibrium initial condition
A, 2 nm
B, 4 nm
C, 20 nm
$\eta = 0.14$
$D_A,\ D_B,\ D_C = 0.1,\ 0.05,\ 0.01\ \mu m^2/s$
start, $\tau = 0$
BD
MPD-RDDME
point particles
$P(n_A)$
occupancy n_A
$P(n_B)$
occupancy n_B
b
MPD-RDDME predicts equilibrium distributions closer to BD
than point-particle simulations
equilibrium, $\tau = 19.8$
equilibrium, $\tau = 19.8$
c
MPD-RDDME approaches the crowded BD diffusivity at long times
mobility of C
$\langle \Delta r_C^2 \rangle$ (nm^2)
time $\tau = D_A t/h^2$
free, $6D_C t$
d
start, $\tau = 0$
equilibrium, $\tau = 20$
bath volume fraction η